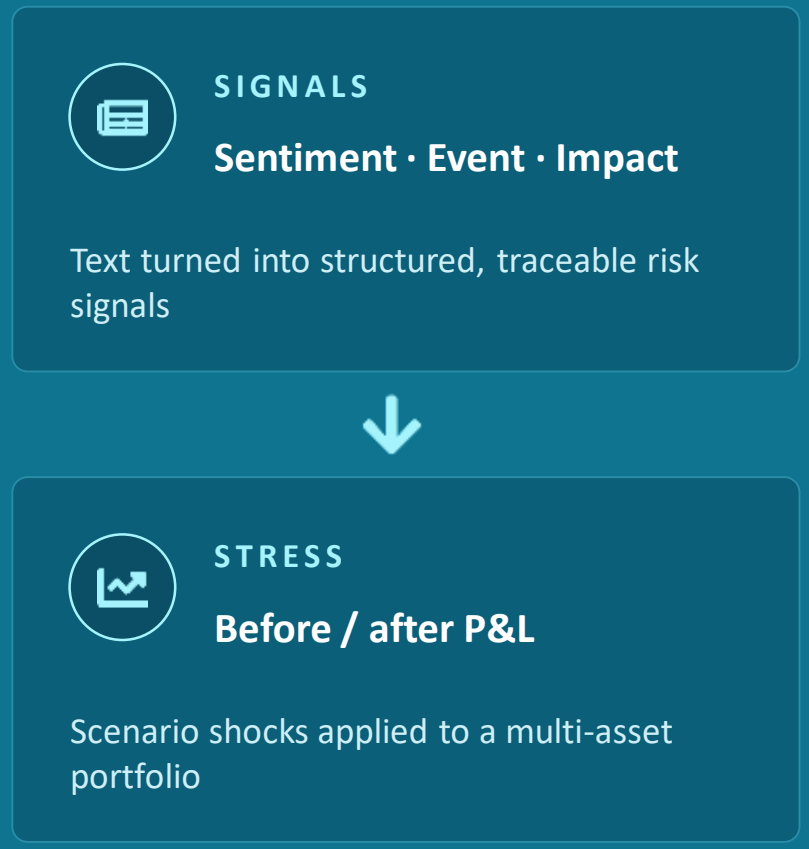
RISK INTELLIGENCE

From market noise to portfolio stress

AI-powered Financial Risk Intelligence & Portfolio Stress Testing

RiskLens converts headlines and social chatter into traceable risk signals, then tests a multi-asset portfolio against configurable shocks.



12
fictional holdings

8
event classes

5
stress scenarios

DEMO DATA AND POSITIONS ARE SYNTHETIC · NOT INVESTMENT ADVICE

Why RiskLens: from unstructured text to decisions

THE CHALLENGE

Market text is fast and unstructured

Hard to see which headlines actually matter for a portfolio.

Scores without provenance are hard to trust

Reviewers need to know where a signal came from and how it was computed.

Risk talk rarely connects to P&L

Qualitative worries seldom turn into a quantified portfolio impact.

THE RISKLENS APPROACH

Structured, traceable risk signals

Sentiment, event type and a 1–10 impact score for every item.

Configurable scenario shocks

Class-level shocks scaled by a user-selected severity.

Transparent portfolio impact

Before/after value, P&L and asset-class attribution.

Traceable · **Configurable** · **Transparent** · **Local-first**

The workflow is designed to be inspectable

01

INGEST

- Bundled samples
- RSS · CSV · JSON
- Synthetic newswire and social pulse

02

ANALYZE

- FinBERT sentiment (optional)
- Deterministic lexicon fallback
- Event classification

03

SIGNAL

- Event type
- Impact score 1–10
- Mapped to candidate scenario

04

STRESS

- Scenario shocks
- Severity multiplier
- Holdings-level P&L

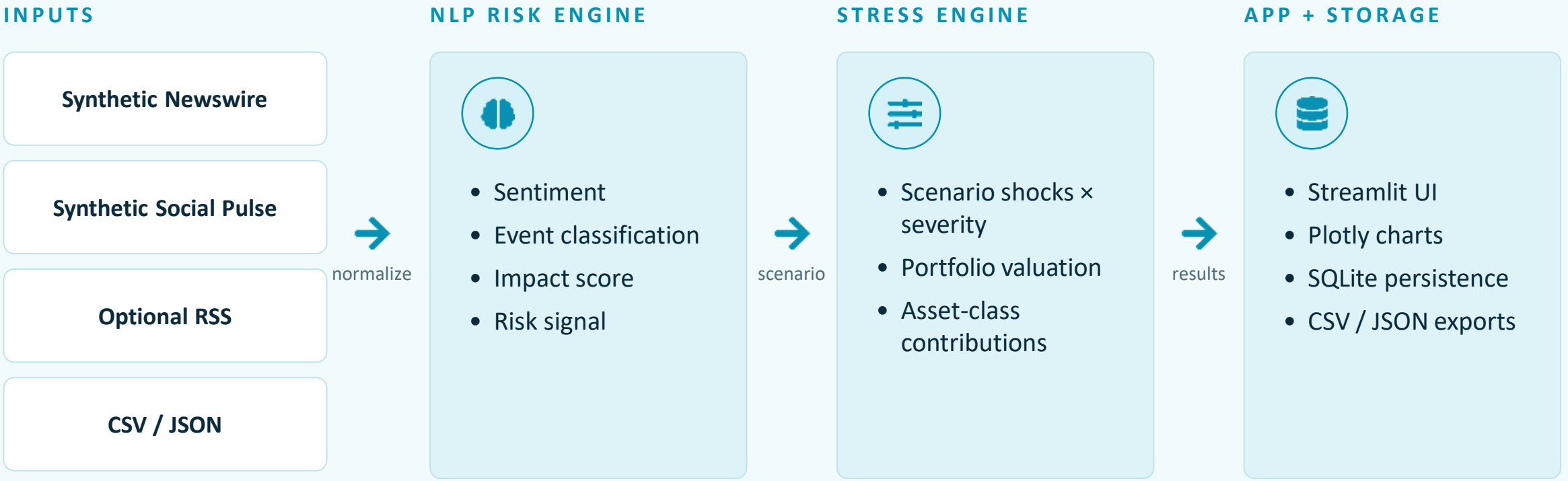
05

REVIEW

- Streamlit + Plotly UI
- SQLite storage
- CSV / JSON exports

Provenance labels and transparent assumptions keep the demo usable even when external APIs or model downloads fail.

Architecture: local-first, modular, reproducible



Local-first

Runs offline on bundled samples; no external API required.

Modular

NLP, stress and app layers are separable components.

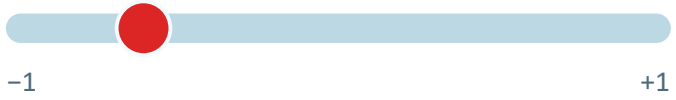
Reproducible

Deterministic fallback and synthetic data give repeatable runs.

NLP: useful signals, explicit limitations

SENTIMENT

Signed score in [−1, +1]



FinBERT optional; the fallback is deterministic and local.

EVENT CLASS

- Geopolitical
- Macroeconomic
- Credit
- M&A
- Earnings
- Regulatory
- Liquidity
- Cyber

IMPACT 1–10

Keyword-based severity proxy



Not a calibrated forecast; signals map to candidate stress scenarios.

ILLUSTRATIVE SIGNAL TRACE

"Regional lender reports rising loan defaults" (synthetic)

Sentiment	Event	Impact	Scenario
−0.62	Credit	8 / 10	Credit crisis

LIMITATION

Model availability is surfaced in each result. The fallback enables offline demos but is not a substitute for validated financial NLP.

Stress lab: quantify scenario exposure

PORTFOLIO

12 fictional holdings

Equities · Bonds · Loans · Derivatives

SCENARIOS

5 configurable shocks

Credit crisis · Rate shock · Geopolitical · Liquidity · Rebound

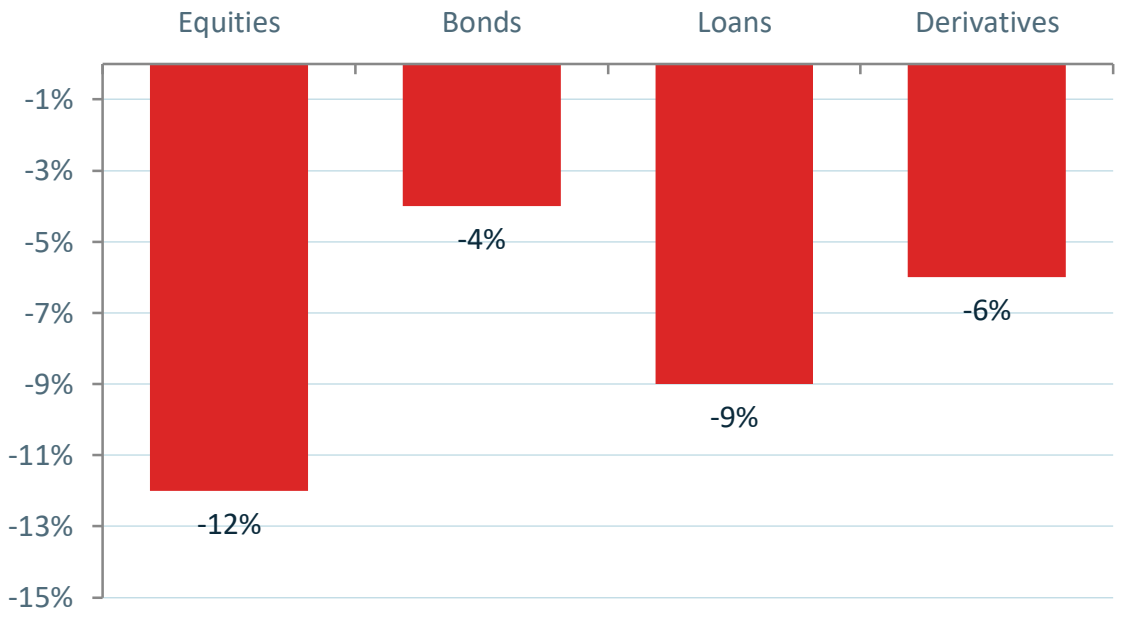
OUTPUTS

Before/after value and P&L

Asset-class attribution · History

Illustrative P&L by asset class

Credit-crisis scenario at severity 1.0 · illustrative values, not model output



Illustrative class-level shocks are multiplied by a user-selected severity. Positions stay static; derivatives use simplified MTM proxies.

Validation, limitations & next steps

VALIDATION

Automated tests cover:

- NLP output bounds
- Event labels
- Portfolio valuation
- Zero-severity invariance
- Risk-signal mapping

LIMITATIONS

Not modeled:

- No historical calibration
- No dynamic hedging
- No transaction costs
- No loan recovery modeling
- No derivative Greeks or live-data guarantees

ROADMAP

Next steps:

1. Calibrate shocks on historical data
2. Benchmark NLP against labeled data
3. Add authenticated data vendors
4. Add model monitoring and audit controls

DEMO TAKEAWAY · Traceable text signal → configurable scenario → transparent portfolio impact

DEMO TAKEAWAY

Inspectable risk intelligence, end to end

Traceable text signal

Sentiment, event type and impact for every headline.

Configurable scenario

Class-level shocks scaled by severity.

Transparent portfolio impact

Before/after P&L with asset-class attribution.

DEMO DATA AND POSITIONS ARE SYNTHETIC · NOT INVESTMENT ADVICE